

1.

$$\text{Power (dBm)} = 10 \log_{10}(P / 1 \text{ mW})$$

$$\text{Power (dBm)} = 10 \log_{10}(10 \text{ W} / 10^{-3} \text{ W}) = 10 \log_{10}(10^4) = 40 \text{ dBm}$$

2.

$$\text{Total Duplex Channels } C = 24 \text{ MHz} / (2 \times 30 \text{ kHz}) = 24,000 / 60 = 400 \text{ channels}$$

(a) Channels per cell (N = 4):

$$C_{\text{cell}} = 400 / 4 = 100 \text{ channels/cell}$$

(b) Maximum offered traffic at 90% capacity:

$$A_{\text{max}} = 0.90 \times 100 = 90 \text{ Erlangs}$$

$$\text{Users/cell } U = 90 / 0.1 = 900 \text{ users/cell}$$

(c) Blocking probability using Erlang B (C = 100, A = 90):

$$P_b = [90^{100} / 100!] / \sum_{k=0}^{100} [90^k / k!] \approx 0.0232 \text{ (2.32\%)}$$

(d) For 120° sectoring (3 sectors/cell):

$$C_{\text{sector}} = 100 / 3 \approx 33 \text{ channels}$$

$$\text{For } C = 33 \text{ and } P_b = 2.32\%, A_{\text{sector}} \approx 26.35 \text{ Erlangs}$$

$$U_{\text{sector}} = 26.35 / 0.1 \approx 263 \text{ users}$$

$$U_{\text{cell}} = 3 \times 263 = 789 \text{ users/cell}$$

(e) Market area = 50 × 50 = 2500 km²; Cells = 2500 / 5 = 500 cells

$$\text{Total subscribers} = 500 \times 900 = 450,000 \text{ subscribers}$$

(f) For 120° sectored antennas:

$$\text{Total subscribers} = 500 \times 789 = 394,500 \text{ subscribers}$$

3.

$$\text{Duplex channel bandwidth} = 2 \times 25 \text{ kHz} = 50 \text{ kHz}$$

(a) Number of duplex channels:

$$C = 20,000 \text{ kHz} / 50 \text{ kHz} = 400 \text{ channels}$$

(b) Channels per cell (N = 4):

$$k = 400 / 4 = 100 \text{ channels/cell}$$

4. Total cells = 84, total channels = 300, $A_u = 0.04$ Erlangs, $P_b = 0.01$ (1%).

(a) Carried traffic ($A_c = A(1 - P_b)$):

$$N = 4: C = 300/4 = 75 \Rightarrow A \approx 62.11 \Rightarrow A_c = 62.11 \times 0.99 = 61.49 \text{ Erlangs/cell}$$

$$N = 7: C = \lfloor 300/7 \rfloor = 42 \Rightarrow A \approx 31.51 \Rightarrow A_c = 31.51 \times 0.99 = 31.19 \text{ Erlangs/cell}$$

$$N = 12: C = 300/12 = 25 \Rightarrow A \approx 16.12 \Rightarrow A_c = 16.12 \times 0.99 = 15.96 \text{ Erlangs/cell}$$

(b) Total system users ($U_{\text{total}} = A/A_u \times 84$):

$$N = 4: U_{\text{cell}} = 62.11/0.04 \approx 1552 \Rightarrow U_{\text{total}} = 1552 \times 84 = 130,368 \text{ users}$$

$$N = 7: U_{\text{cell}} = 31.51/0.04 \approx 787 \Rightarrow U_{\text{total}} = 787 \times 84 = 66,108 \text{ users}$$

$$N = 12: U_{\text{cell}} = 16.12/0.04 \approx 403 \Rightarrow U_{\text{total}} = 403 \times 84 = 33,852 \text{ users}$$

5.

(a) Raw data rate per user:

$$R_{\text{raw}} = 270.833 \text{ kbps} / 8 = 33.854 \text{ kbps}$$

(b) Traffic efficiency:

$$\text{Useful rate} = 33.854 - 10.1 = 23.754 \text{ kbps}$$

$$\text{Efficiency} = (23.754 / 33.854) \times 100\% \approx 70.16\%$$

6.

(a) Total channels available = 20 (numbered 1 through 20).

(b) Channel assignment strategies:

- Fixed Channel Assignment (FCA): Channels are divided into fixed sets per cluster. For $N = 4$, each group gets 5 channels; e.g., Group 1 gets {1, 5, 9, 13, 17}. Co-channel cells reuse the exact same set.
- Dynamic Channel Assignment (DCA): Channels are not permanently assigned to specific cells. All 20 channels stay in a central pool and are assigned on demand based on traffic and interference conditions.

7. DECT (Digital Enhanced Cordless Telecommunications):

Frequency band: 1880-1900 MHz (Europe/Global) and 1920-1930 MHz (US DECT 6.0).

Key features:

- Dynamic Channel Selection (DCS): Automatically selects the clearest channel to avoid interference.
- High security: Uses DSAA authentication and DSEC voice encryption.
- Dedicated spectrum: Prevents interference from Wi-Fi networks.
- Seamless handoff: Allows users to roam between cells without losing connection.

8. Given: Total cells = 32, R = 1.6 km, total channels = 336, N = 7.

$$\text{Cell area} = (3\sqrt{3}/2)R^2 \approx 2.598 \times (1.6)^2 \approx 6.65 \text{ km}^2$$

$$\text{Total area covered} = 32 \times 6.65 \text{ km}^2 \approx 212.8 \text{ km}^2$$

$$\text{Channels per cell} = 336 / 7 = 48 \text{ channels/cell}$$

$$\text{Total concurrent calls} = 32 \times 48 = 1536 \text{ concurrent calls}$$

9. ISDN types:

BRI (Basic Rate Interface): Structure: 2B + 1D channels (2 × 64 kbps + 16 kbps). Speed: 144 kbps. Uses: residential internet, small office lines, backup connections.

PRI (Primary Rate Interface): Structure: T1 (23B + 1D) or E1 (30B + 1D).

Speed: 1.544 Mbps (T1) / 2.048 Mbps (E1). Uses: large enterprise PBX trunk lines, digital call centers.

B-ISDN (Broadband ISDN): Structure: ATM-based packet switching.

Speed: 155 Mbps to 622 Mbps+. Uses: high-speed telecommunication network backbones.

10.

$$\text{Number of radio channels} = 20 \text{ MHz} / 20 \text{ kHz} = 20,000/20 = 1000 \text{ radio channels}$$

$$\text{Simultaneous users} = 1000 \times 6 = 6000 \text{ simultaneous users}$$

11. SS7 (Signaling System No. 7):

Functions: Out-of-band signaling system that handles call setup, routing, teardown, SMS, and roaming lookup (HLR/VLR).

Call setup steps:

- IAM (Initial Address Message): Requests circuit reservation along the call path.
- ACM (Address Complete Message): Confirms the line is ringing at the destination.
- ANM (Answer Message): Connects the voice trunk when the called party picks up.
- REL / RLC (Release / Release Complete): Releases the circuit when the call ends.

Advantages over in-band signaling: Faster setup time; voice circuits are not blocked during setup; prevents fraud such as tone-manipulation attacks.

12. Call rate $\lambda = 1800$ calls/hr, holding time $H = 3$ min = $3/60$ hr = 0.05 hr.

$$\text{Traffic intensity } A = \lambda \times H = 1800 \times 0.05 = 90 \text{ Erlangs}$$

13. Total bandwidth = 33 MHz = 33,000 kHz; duplex channel bandwidth = 2×25 kHz = 50 kHz.

$$C = 33,000 / 50 = 660 \text{ duplex channels}$$

$$\text{For } N = 4: k = 660/4 = 165 \text{ channels/cell}$$

$$\text{For } N = 7: k = \lfloor 660/7 \rfloor = 94 \text{ channels/cell}$$

$$\text{For } N = 12: k = 660/12 = 55 \text{ channels/cell}$$

14.

$$\text{Raw data rate per user} = 48.6 \text{ kbps} / 3 \text{ users} = 16.2 \text{ kbps}$$